Four Thesis-Driven Alpha Families

From economic thesis to submitted alpha:
construction, upgrades and the best alpha in each family

- Closing-pressure reversal
- Gap-down rebound
- Volatile-close reversal
- Slow accounting composite

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Trexquant Alpha Generation Competition 2026 · Trexsim (US equities, delay 0, Slow mode)

Summary: four families, four best alphas

Each family starts from one economic thesis; the highlighted alpha is the best of its family

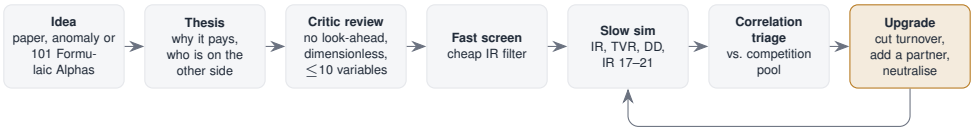
Family	Thesis (one line)	Best alpha	IR	TVR	DD	IR/$\sqrt{\text{TVR}}$	IR 17–21
A. Closing-pressure reversal	Late-day, price-insensitive flow pushes the close away from VWAP; the push reverts	**#708808**	**.118**	.624	**6.2**	.149	**.145**
B. Gap-down rebound	A close pinned at the low after a gap-down is forced selling; liquidity providers are paid	**#724045**	**.129**	.953	11.5	.132	.075
C. Volatile-close reversal	A close noisier than the open signals flow, not news, so the last-hour move fades	**#685104**	.092	.328	12.0	.161	.062
D. Slow accounting composite	Many weak, slow, low-correlated accounting premia add up to a steady signal	**#716550**	.077	**.036**	12.4	**.407**	**.139**

- All four best alphas are **submitted** and qualify on the portal.
- Pairwise daily-PnL correlation between them: **0.06 to 0.48**.

IR = daily IR ($\times 15.9$ for annual). TVR = daily turnover. DD = max drawdown (%). IR 17–21 = IR over 2017–2021, the most recent 5 years. **Best in family** = highest recent (2017–21) IR among the family's submitted alphas, i.e. the most robust one.

Research process

Thesis first; every alpha is a small, commented program with a written rationale and a source



What we learned

- Most daily IR in this universe comes from **short-term reversal**, so pure reversal ideas collide with the pool (correlation .78–.99).
- Fast and Slow PnL of the same alpha correlate only **0.34**: we only trust Slow results.
- The fix that worked: keep the thesis signal and add a **slow partner whose returns move against reversal** (low accruals: daily PnL correlation ≈ -0.2). That cuts turnover and crowding without killing IR.

By the numbers

- **403** alphas simulated, **10** submitted.
- 4 families, each traced below from the first version to the best alpha.
- Every alpha documents its thesis, source and a comment on each line of code.

A. Closing-pressure reversal: thesis

Fade the gap between the close and the last-hour VWAP when the close was unusually busy

Thesis

- Index rebalancing, end-of-day hedging and fund flows trade **price-insensitively** in the last hour.
- This pushes the close away from the hour's VWAP. Nobody is trading on news, so the push **reverts** over the next days.
- The effect should be strongest on **busy closes** (many trades vs. the stock's own 20-day norm).

Why add low accruals

- Cash-backed earnings (Sloan) earn a slow premium whose daily PnL moves *against* reversal, so the blend keeps the reversal IR with lower turnover and less overlap with crowded reversal alphas.

Sources: Campbell, Grossman & Wang (QJE 1993); Bogousslavsky & Muravyev (2023), closing-auction price pressure; Sloan (The Accounting Review 1996).

Best alpha #708808 (simplified)

```
g    = ts_mean((close - vwap_last_hour)/vwap_last_hour, 5)
busy = cs_rank(trades_last_hour/ts_mean(trades_last_hour,20)) > 1.3
p1   = -cs_rank(g) * busy              # fade the gap on busy closes
hot  = cs_rank(volume/ts_mean(volume,20)) > 1.3
p2   = -(cs_rank(ts_mean(ret_last_hour,5)) + cs_rank(g)) * hot
rev  = cs_factorneut(p1 + p2, [size, volatility])
acc  = -cs_rank((net_income - cash_from_ops)/total_assets)
alpha = rev + 1.6 * acc                # reversal + low accruals
```

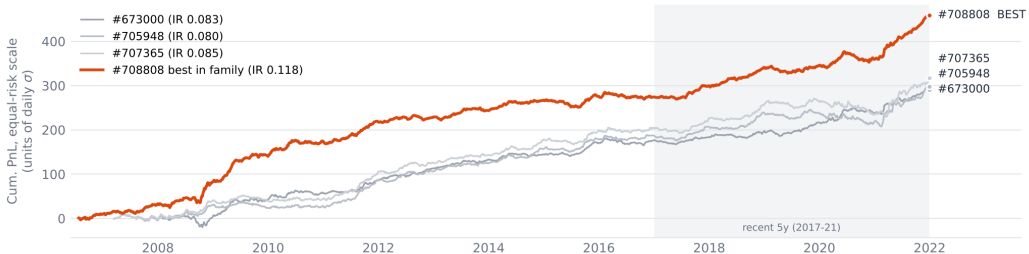
First version #673000: the pure thesis

```
g    = ts_mean((close - vwap_last_hour)/vwap_last_hour, 5)
busy = cs_rank(trades_last_hour/ts_mean(trades_last_hour,20)) > 1.3
alpha = -cs_rank(g) * busy
```

A. Closing-pressure reversal: from thesis to best alpha

IR .083 → .118, drawdown halved (12.5 → 6.2), recent IR .085 → .145

	Alpha	Change	IR	TVR	DD	IR/$\sqrt{\text{TVR}}$	IR 17–21
1	#673000	Pure thesis: fade close-vs-VWAP gap on busy closes. Pool correlation .78	.083	1.047	12.5	.081	.085
2	#705948	+ R&D, buybacks, low accruals; size-neutral; book held for the week	.080	.248	11.7	.161	.069
3	#707365	Accruals weight 1.5× (submitted)	.085	.226	11.6	**.179**	.077
4	**#708808** BEST	Two late-day pressure legs, size/vol-neutral, + 1.6× low accruals (submitted)	**.118**	.624	**6.2**	.149	**.145**



Curves traced from the Trexsim PnL charts and scaled to equal risk, so a steeper line means a higher IR. Shaded: the recent 5 years (2017–21) used to pick the best alpha.

B. Gap-down rebound: thesis

Buy stocks that opened down and closed pinned at the day's low (Alpha #54, adapted)

Thesis

- A stock that **gaps down** and closes **at its low** was sold all day: margin calls, stop-losses, forced selling.
- Short-term reversal is the return to **providing liquidity**; it is largest when sellers are impatient.
- The $(\text{open}/\text{close})^5$ term scales the bet by the size of the drop.

Why add a slow partner

- The raw formula is public: our standalone version was **.995** correlated with an alpha already in the pool.
- A slow balance-sheet partner (short-term liabilities/assets) changes the PnL path and gave the best recent IR.

Sources: Kakushadze, "101 Formulaic Alphas" (2016), arXiv:1601.00991; Nagel, "Evaporating Liquidity" (RFS 2012); Sloan (1996).

Best alpha #724045

```
loc  = (low - close)/(low - high - 0.001*close)  # 0 = closed at low
b    = cs_zscore(-loc * (open/close)**5)          # Alpha #54
p_cl = cs_zscore(cs_rank(current_liabilities/total_assets))
alpha = b + 1.5 * p_cl
```

First submission #715966

```
acc  = cs_zscore(-cs_rank((net_income - cash_from_ops)/total_assets))
alpha = b + 1.5 * acc                            # low accruals partner
```

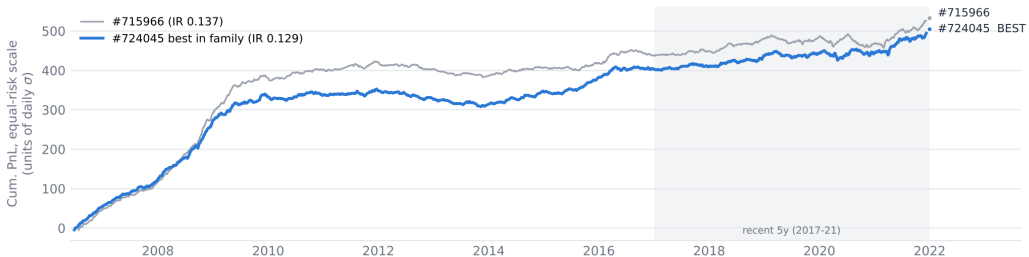
First version (standalone Alpha #54)

```
alpha = -loc * (open/close)**5
```

B. Gap-down rebound: from thesis to best alpha

Highest IR of all families; recent IR .015 → .075 after adding slow partners

	Alpha	Change	IR	TVR	DD	IR/$\sqrt{\text{TVR}}$	IR 17–21
1	Alpha #54	Standalone formula. Pool correlation .995: already taken	.112	1.381	–	.095	.015
2	#715966	+ 1.5× low accruals (submitted)	**.137**	.994	**10.6**	**.137**	.064
3	**#724045** BEST	Partner from grid search: current liabilities/assets at 1.5× (submitted)	.129	.953	11.5	.132	**.075**



Curves traced from the Trexsim PnL charts and scaled to equal risk, so a steeper line means a higher IR. Shaded: the recent 5 years (2017–21) used to pick the best alpha.

Honest read: most PnL came in 2006–09, when liquidity provision pays most. #724045 beats #715966 on recent IR and pool correlation (.77 vs .83); the two are .84 correlated.

C. Volatile-close reversal: thesis

Fade the last-hour move only where the close was more volatile than the open

Thesis

- News is absorbed at the **open**; a close that is **noisier than the open** is driven by flow (hedging, rebalancing, liquidations), not information.
- Flow-driven moves revert. So we fade the 5-day last-hour return only for stocks whose last hour is unusually volatile relative to their first hour.
- This works best when markets are stressed: a **crisis alpha** that diversifies the other families.

Why add slow partners

- Less common slow premia (stock-based pay, buybacks, free-cash-flow yield, low accruals) carry most of the risk, so the mix differs from crowded reversal and fundamental alphas.

Sources: Bogousslavsky & Muravyev (2023); Nagel (RFS 2012); Pontiff & Woodgate (JF 2008); Lakonishok, Shleifer & Vishny (JF 1994); Sloan (1996).

Best alpha #685104 (simplified)

```
late = cs_rank(ts_mean(std_ret_last_hour /
               (std_ret_first_hour + 1e-4), 5)) > 1.5
r  = -cs_rank(ts_mean(ret_last_hour, 5)) * late  # fade flow
k1 = cs_rank(stock_based_comp / mktcap)          # intangibles
k2 = -cs_rank(shsout/ts_delay(shsout,250) - 1)   # buybacks
k3 = cs_rank(free_cash_flow / mktcap)            # FCF yield
k4 = -cs_rank((net_income - cash_from_ops)/total_assets)
alpha = r + 2 * (k1 + k2 + k3 + k4)
```

Upgrade path

```
#678911  r + 2*(k1+k2+k3)       first version
#684336  r + 3*(k1+k2+k3)       our first submission
#685104  r + 2*(k1+k2+k3+k4)    + low accruals  -> BEST
```

C. Volatile-close reversal: from thesis to best alpha

Drawdown 17.7 → 12.0, turnover .54 → .33, IR .075 → .092

	Alpha	Change	IR	TVR	DD	IR/$\sqrt{\text{TVR}}$	IR 17–21
1	#678911	Volatile-close reversal + stock pay, buybacks, FCF yield at 2×	.075	.540	17.7	.102	.034
2	#684336	Slow weight 3× (our first submission)	.073	.486	17.7	.105	.031
3	**#685104** BEST	+ low accruals as 4th slow leg, 2× (submitted)	**.092**	**.328**	**12.0**	**.161**	**.062**



Curves traced from the Trexsim PnL charts and scaled to equal risk, so a steeper line means a higher IR. Shaded: the recent 5 years (2017–21) used to pick the best alpha.

Honest read: gains cluster in stressed markets (2008, 2011, 2020–21) and are flat in calm 2014–19, as the crisis-alpha thesis predicts.

D. Slow accounting composite: thesis

Weak, slow, low-correlated balance-sheet premia combined into one steady signal

Thesis

- Each accounting ratio (item/assets) is a **weak** predictor (IR .03–.05) but changes only quarterly, so turnover is tiny.
- The ratios are **weakly related**, so averaging diversifies their noise: higher IR at the same turnover, so a very high IR/$\sqrt{\text{TVR}}$.
- Long cash-backed earnings, operating-liability funding, older asset bases; short goodwill-heavy (acquisitive) firms.

Guarding against data-mining

- Every leg's sign was checked to hold **out of sample (2017–21)** before it was included.

Sources: Sloan (1996); Green, Hand & Zhang (RFS 2017); Cooper, Gulen & Schill (JF 2008); Pontiff & Woodgate (JF 2008); Novy-Marx (JFE 2013).

Best alpha #716550

```
ta = ts_fill(total_assets)
cl = cs_rank(current_liabilities / ta)      # operating funding
ar = cs_rank(accounts_receivable / ta)      # receivables
gw = -cs_rank(goodwill / ta)                # short acquirers
ad = cs_rank(accumulated_depreciation / ta) # older assets
oc = cs_rank(other_current_assets / ta)
acc = -cs_rank((net_income - cash_from_ops) / ta)
alpha = cl + ar + gw + ad + oc + acc        # equal weight
```

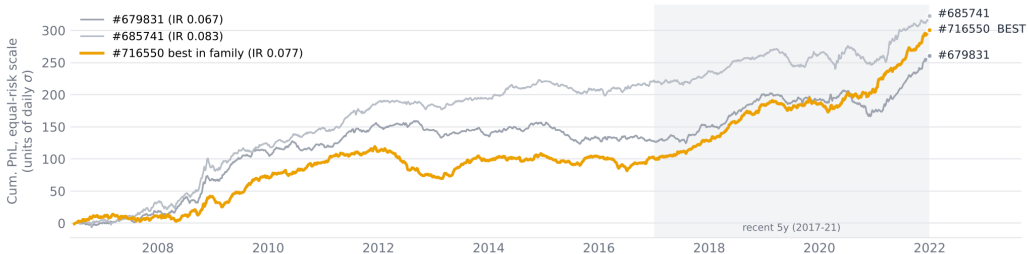
Upgrade path

```
#679831  low accruals + low asset growth          (2 premia)
#685741  7 classic premia: accruals, issuance,
         FCF, asset growth, R&D, gross profit, SBC
#716550  balance-sheet composition + accruals  -> BEST
```

D. Slow accounting composite: from thesis to best alpha

IR/$\sqrt{\text{TVR}}$ .30 → .41, turnover .036, strongest recent performance of all families

	Alpha	Change	IR	TVR	DD	IR/$\sqrt{\text{TVR}}$	IR 17–21
1	#679831	Low accruals + low asset growth	.067	.051	11.2	.298	.095
2	#685741	7-premium composite (submitted)	**.083**	.051	**11.1**	.366	.076
3	**#716550** BEST	Balance-sheet composition (5 items/assets) + low accruals (submitted)	.077	**.036**	12.4	**.407**	**.139**

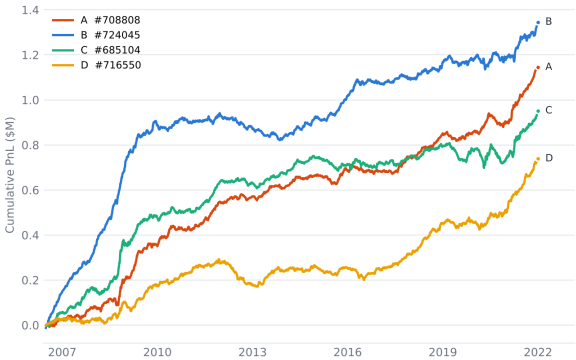


Curves traced from the Trexsim PnL charts and scaled to equal risk, so a steeper line means a higher IR. Shaded: the recent 5 years (2017–21) used to pick the best alpha.

#716550 beats #685741 (higher full-period IR) on recent IR (almost 2×), turnover and pool correlation (.53 vs .79).

Diversification across families and takeaways

The four best alphas make money at different times



Cumulative PnL in $M of the four best alphas, 2006–2021 (traced from the Trexsim PnL charts).

Daily PnL correlation (2020–21)

	A	B	C	D
A #708808	1	.20	.48	.25
B #724045	.20	1	.06	.19
C #685104	.48	.06	1	.17
D #716550	.25	.19	.17	1

Takeaways

- Every best alpha starts from a **clear economic thesis** with a named counterparty.
- The upgrade that worked: keep the thesis signal, add a **slow partner** that moves against reversal; it cuts turnover and crowding.
- The four families are **largely uncorrelated** (max .48), so together they form a diversified book.

Code, theses and the full log of 403 simulations:
`https://github.com/abhinavteja2005/trexquant-alpha-research`